

# Introduction to Computing at Scale

The Design, Operation, and Societal Impacts of Data Centers

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## Information & Computing

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# What is Information?

*“Knowledge obtained from investigation, study, or instruction”*

— Merriam-Webster

## **Digital**

Binary data in memory

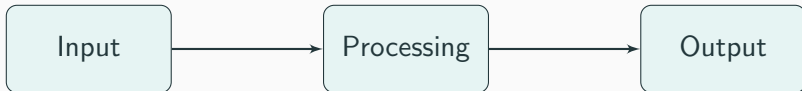
## **Algorithmic**

Output of an algorithm

## **Physical**

Heat from a stove

# Information Processing is Valuable



- **Biology:** Hunger → Look for food → Eat → Not hungry.
- **Umwelt:** How different species experience the world.
- *Recommended Reading:* *An Immense World* by Ed Yong.

Humans have intuition (more vs. less or nothing), but precision and collaboration are needed.

## The Solution: MATH

Math serves as a common framework.

- **Standards** → Easier communication.
- Number systems, operations, written forms.

**Computation** = Information Processing + Math?

# Why We Need Computers

The human brain is **bad at math**, and we aren't meant to be tools.

- **Limited Memory:** Working memory of 5–9 'numbers'
- **Unreliable:** Forgetful and non-operational for long periods of time (sleep)
- **Imprecise**
- **Hard to scale**
- **Slow:** Long training times (childhood/education).

*We needed an optimized tool for the job.*

**Computing** = Information Processing using Math?

**Definitions:**

- To determine especially by mathematical means
- To determine or calculate by means of a computer

# Computing Hardware

How do we build fast, small, and efficient computers?

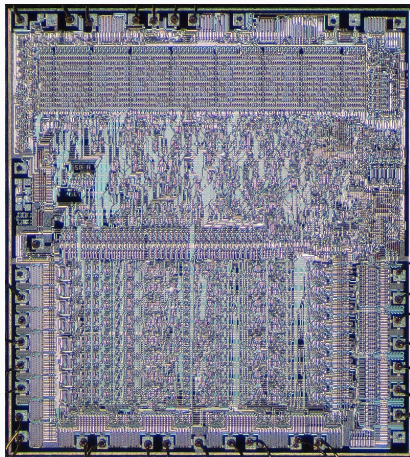
Currently and for many decades:

## Transistors

- **Digital:** 0s and 1s.
- **Electrical:** Voltage levels.

*Although this doesn't mean there aren't other alternatives*

*(see quantum, analog, neuro, etc).*



# Computing at Scale

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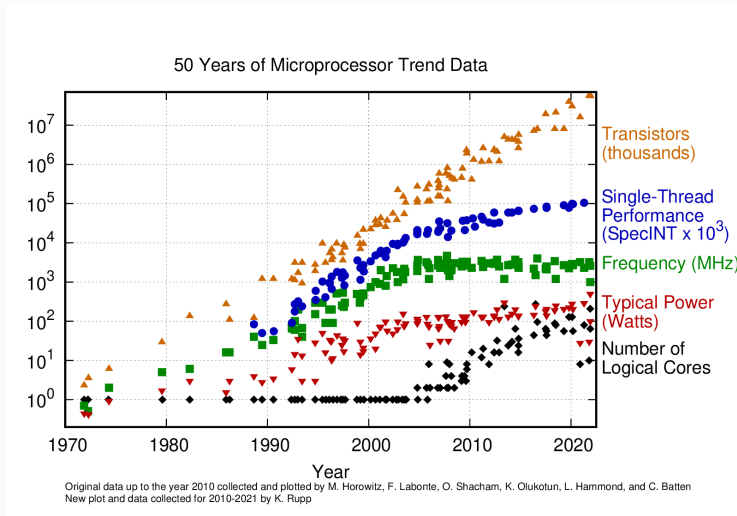
**Moore's Law:** Transistor count per microprocessor doubles every  $\approx 2$  years.

**Dennard Scaling:**

- Power density stays constant as transistors shrink.
- Same chip area  $\rightarrow$  Same power draw.

*These trends drove decades of progress.*

# 50 Years of Trends

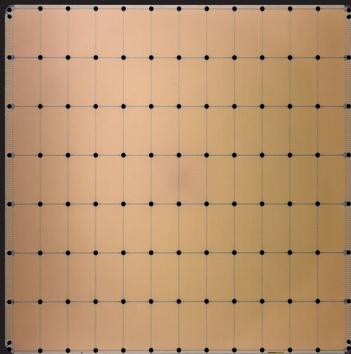


**Figure 1:** Transistors, Frequency, Power, and Cores (1970–2020)

# Wafer-Scale Engines

Making bigger chips. From reticle-sized to wafer-sized.

## Cerebras Wafer-Scale Engine



**Cerebras WSE-2 7nm**  
2.6 Trillion Transistors  
46,225 mm<sup>2</sup> Silicon

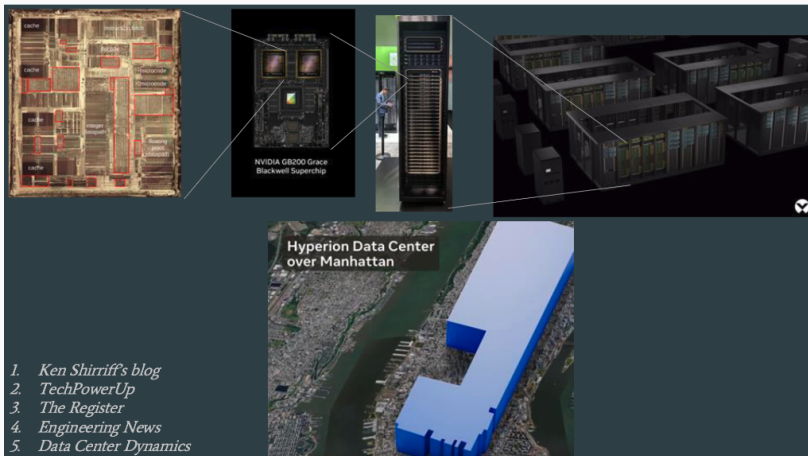


**Largest GPU**  
54.2 Billion Transistors  
826 mm<sup>2</sup> Silicon

# Beyond the Chip

Scale isn't just on-chip. It's the whole system.

Transistors → Chips → Boards → Racks → Aisles → Datacenters.



# The Scales of Computing

We scaled transistors **down**, and hit hard physical limits.

We scaled **up** and made bigger chips.

We connected more chips together and scaled **out**.

Now even a single datacenter is too small, and we scale **across**.

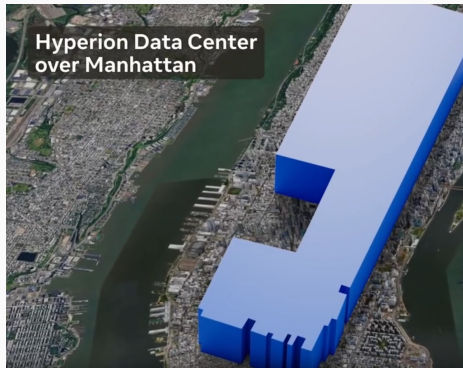
## Impact: Power & Water

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# Unprecedented Energy Scale

## The Jump in Scale

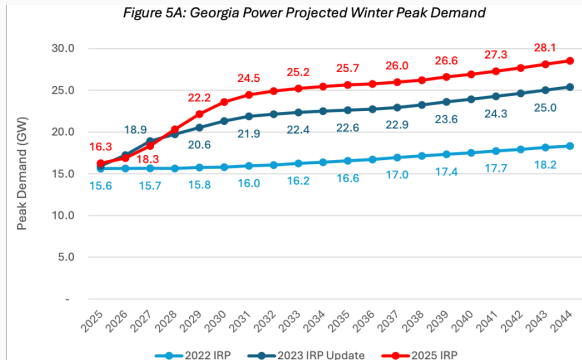
- **Past:** Top supercomputers  $\approx 20$  MW
- **Future:** Meta's "Hyperion" DC  $\approx 5$  GW.
- Note: A nuclear plant outputs  $\approx 1$  GW.



Hyperion Data Center scale over Manhattan

# The Strain on our Energy Grid.

- Demand is outstripping generation capacity.
- Demand spiked **rapidly**. Decadal planning → ASAP.
- AI training loads can fluctuate 100s of MWs in seconds.
- Inflated demand through speculation and multiple bids
- AI bubble?



Source: Georgia Power IRP

# The Water Cost

**Massive Power + Inefficiencies = Cooling Demand**

## The Cost of Evaporative Cooling

- Evaporative cooling is very effective and power-efficient, but not without tradeoffs.
- A single DC *can* evaporate millions of gallons of water/day.
- Equivalent to the usage of hundreds of thousands of people.
- Can be mitigated by: free-air/dry/adiabatic cooling, water reuse, etc.

**Secondary Water Impact:** Thermoelectric power generation is already the largest water consumer in the US.

# The Future

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# In Space and Under the Sea

We're reconsidering the environments computing operates in.

Both extremes are being actively tested.

## Space

- Solar energy availability
- Cooling challenges
- Radiation/error tolerance
- Cost of deployment and repairs
- Inter-satellite communication

## Sea

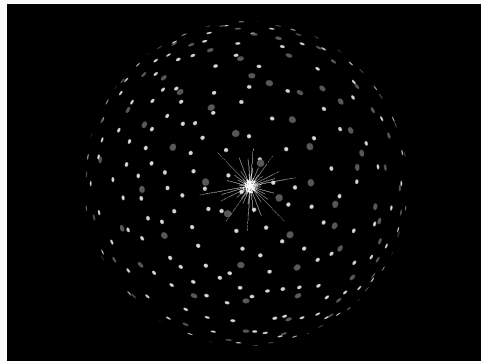
- Water availability
- Environmental effects
- Cost of deployment and repairs

# Limits of Scale/Computing

What is the limit of computing at scale?

Is there a limit?

Should we self-impose a limit?



Artistic depiction of a Matrioshka Brain

# Efficiency and Jevons Paradox

- **The Efficiency Trend (Koomey's Law):**
  - Computations per joule doubled every  $\approx 1.57$  years for decades (now slower).
  - *Result:* Modern computers are vastly more efficient than their predecessors.
- **Jevons Paradox:**
  - Efficiency improvements induce increased demand and overall resource consumption.
  - Gains in computing efficiency unlock new applications and demand (e.g. we now use AI for internet searches).

## The Divergence

- **Lighting Demand:** Capped by biology/physics (e.g., the human eye).
- **Computing Demand:** Capped only by available energy and heat dissipation?

**Will gains in computing efficiency lead to reduced resource consumption?**

# Sufficiency, Sobriety, and Ethics

**Sufficiency:** What is the least amount of resources/computing to complete a task?

- Example: Can I use a linear regression or do I need a 1 trillion parameter model?
- **Self**-sufficiency example: Solar-powered pocket calculator

**Sobriety:** Is this a problem I need (computing) to solve?

- Conscious evaluation of tech use. **Re-examine our relationship with technology.**
- Example: Does my toaster need AI to make my morning toast?

**Ethics:** How do we ensure computing systems at scale are ethically designed, built, operated, used, and decommissioned?

## Summary: Computing at Scale

The full stack of computing at scale:

1. **The Foundation:** Information processing is valuable, and we use physical computers to perform this task.
2. **The Engine:** Transistors drove 50 years of exponential growth (Moore/Dennard), but are hitting lower limits.
3. **The New Scale:** We now scale up, out, and across to form massive systems (datacenters).
4. **The Constraint:** This massive scale is colliding with physical realities: energy grids, water supplies, and communities (us).

Our technical ability to scale is **limitless**,  
but our resources and ethical boundaries are **limited**.

**What do you want the future  
of computing to look like?**